Instr	OPCode	*ALUOP(4b)*	*ALUSrcSel(2b)*	*RegWrite(1b)*	*AdvMemRead(1b)*	*MemRead(1b)*	*MemWrite(1b)*	*PCStackWrite(1b)*	*PcSrc(2b)*	*PCStackOP(1b)*	*CondJump(2b)*
NOP	b0000	x	x	0	x	x	0	0	b00	x	x
ADD	b0001	b0000	b00	1	0	0	0	0	b00	x	x
ADC	b0001	b0001	b00	1	0	0	0	0	b00	x	x
SUB	b0001	b0010	b00	1	0	0	0	0	b00	x	x
AND	b0001	b0011	b00	1	0	0	0	0	b00	x	x
OR	b0001	b0100	b00	1	0	0	0	0	b00	x	x
XOR	b0001	b0101	b00	1	0	0	0	0	b00	x	x
SLL	b0001	b0110	b00	1	0	0	0	0	b00	x	x
SRL	b0001	b0111	b00	1	0	0	0	0	b00	x	x
CMP	b0001	b1000	b00	1	0	0	0	0	b00	x	x
MOV	b0001	b1001	b00	1	0	0	0	0	b00	x	x
ADD&	b0001	b0000	b10	0	0	1	1	0	b00	x	x
SUB&	b0001	b0001	b10	0	0	1	1	0	b00	x	x
STORE	b0001	b1001	b10	0	0	x	1	0	b00	x	x
LOAD	b0001	x	b0x	1	0	1	0	0	b00	x	x
ADDI	b0010	b0000	b01	1	0	0	0	0	b00	x	x
SUBI	b0011	b0010	b01	1	0	0	0	0	b00	x	x
ADDI&	b0100	b0000	b11	0	0	1	1	0	b00	x	x
SUBI&	b0101	b0010	b11	0	0	1	1	0	b00	x	x
MOVI	b0110	b1001	bx1	1	0	0	0	0	b00	x	x
MOVI&	b0111	b1001	b11	0	0	x	1	0	b00	x	x
LDRAM	b1000	x	x	1	1	1	0	0	b00	x	x
STORAM	b1001	x	x	1	1	1	1	0	b00	x	x
J	b1010	x	x	0	x	x	0	0	b10	x	b00
JEQ	b1011	x	x	0	x	x	0	0	b10	x	b01
JNE	b1100	x	x	0	x	x	0	0	b10	x	b10
JLT	b1101	x	x	0	x	x	0	0	b10	x	b11
CALL	b1110	x	x	0	x	x	0	1	b10	0	b00
RET	b1111	x	x	0	x	x	0	1	b01	1	x